Infinite Series Worksheet (9.3) — Solutions

Type A: Geometric Series

$\sum_{k=0}^{\infty} ar^k = a + ar + ar^2 + \cdots$ converges to $\dfrac{a}{1-r}$ if $|r| < 1$ and diverges if $|r| \geq 1$. Here a is always the *first term* and r is the ratio of consecutive terms.

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1. $\displaystyle\sum_{k=0}^{\infty} \frac{7}{3^k}$

Geometric with $a = 7$ and $r = \frac{1}{3}$. Since $|r| = \frac{1}{3} < 1$, the series converges and $\dfrac{a}{1-r} = \dfrac{7}{1-\frac{1}{3}} = \dfrac{7}{2/3} = \dfrac{21}{2}$.

Answer: converges, sum $= \frac{21}{2}$

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2. $\displaystyle\sum_{k=0}^{\infty} \frac{(-1)^k\, 6}{5^k}$

Rewrite as $\sum_{k=0}^{\infty} 6\left(-\frac{1}{5}\right)^k$: geometric with $a = 6$, $r = -\frac{1}{5}$. Since $|r| = \frac{1}{5} < 1$, it converges and $\dfrac{a}{1-r} = \dfrac{6}{1+\frac{1}{5}} = \dfrac{6}{6/5} = 5$.

Answer: converges, sum $= 5$

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3. $\displaystyle\sum_{k=1}^{\infty} 2^{2k}\, 3^{1-k}$

Concealed form: $\displaystyle\sum_{k=1}^{\infty} 2^{2k}\, 3^{1-k} = \sum_{k=1}^{\infty} \frac{4^k}{3^{k-1}} = \sum_{k=1}^{\infty} 4\left(\frac{4}{3}\right)^{k-1}$. Since $r = \frac{4}{3} > 1$, the series diverges.

Answer: diverges

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4. $\displaystyle\sum_{k=1}^{\infty} \frac{5^{k+1}}{8^k}$

$\dfrac{5^{k+1}}{8^k} = \dfrac{5^2 \cdot 5^{k-1}}{8 \cdot 8^{k-1}} = \dfrac{25}{8}\left(\dfrac{5}{8}\right)^{k-1}$, so $a = \frac{25}{8}$ and $r = \frac{5}{8}$. Since $|r| < 1$, it converges and $\dfrac{a}{1-r} = \dfrac{25/8}{3/8} = \dfrac{25}{3}$.

Answer: converges, sum $= \frac{25}{3}$

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5. $\displaystyle\sum_{k=2}^{\infty} \frac{3^k}{2^{3k-1}}$

$\dfrac{3^k}{2^{3k-1}} = \dfrac{2 \cdot 3^k}{8^k} = 2\left(\dfrac{3}{8}\right)^k$. The first term ($k = 2$) is $a = \frac{9}{32}$ and $r = \frac{3}{8}$, $|r| < 1$, so $\dfrac{a}{1-r} = \dfrac{9/32}{5/8} = \dfrac{9}{32} \cdot \dfrac{8}{5} = \dfrac{9}{20}$.

Answer: converges, sum $= \frac{9}{20}$

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6. $3 + \dfrac{3}{4} + \dfrac{3}{4^2} + \cdots + \dfrac{3}{4^{k-1}} + \cdots$

This is $\sum_{k=1}^{\infty} 3\left(\frac{1}{4}\right)^{k-1}$: geometric with $a = 3$, $r = \frac{1}{4}$. Since $|r| < 1$, it converges and $\dfrac{a}{1-r} = \dfrac{3}{3/4} = 4$.

Answer: converges, sum $= 4$

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7. $\dfrac{2}{9} - \dfrac{4}{27} + \dfrac{8}{81} - \cdots + \dfrac{(-1)^{k-1}\, 2^k}{3^{k+1}} + \cdots$

$\dfrac{(-1)^{k-1}2^k}{3^{k+1}} = \dfrac{2}{9}\left(-\dfrac{2}{3}\right)^{k-1}$ (check: each term is $-\frac{2}{3}$ times the previous one), so $a = \frac{2}{9}$, $r = -\frac{2}{3}$, $|r| < 1$: $\dfrac{a}{1-r} = \dfrac{2/9}{5/3} = \dfrac{2}{15}$.

Answer: converges, sum $= \frac{2}{15}$

Type B: Telescoping Sums

Write s_n in closed form, then $\sum_{k=1}^{\infty} u_k = \lim_{n\to+\infty} s_n$ (if the limit exists).

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8. $\displaystyle\sum_{k=1}^{\infty} \frac{1}{k^2+3k+2}$

$\frac{1}{(k+1)(k+2)} = \frac{1}{k+1} - \frac{1}{k+2}$, so $s_n = \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \cdots + \left(\frac{1}{n+1} - \frac{1}{n+2}\right) = \frac{1}{2} - \frac{1}{n+2} \longrightarrow \frac{1}{2}$.

Answer: converges, sum $= \frac{1}{2}$

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9. $\displaystyle\sum_{k=1}^{\infty} \frac{1}{(2k-1)(2k+1)}$

$\frac{1}{(2k-1)(2k+1)} = \frac{1}{2}\left(\frac{1}{2k-1} - \frac{1}{2k+1}\right)$, so $s_n = \frac{1}{2}\left[\left(1 - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \cdots + \left(\frac{1}{2n-1} - \frac{1}{2n+1}\right)\right] = \frac{1}{2}\left(1 - \frac{1}{2n+1}\right) \to \frac{1}{2}$.

Answer: converges, sum $= \frac{1}{2}$

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10. $\displaystyle\sum_{k=1}^{\infty} \ln\left(\frac{k}{k+1}\right)$

$\ln\left(\frac{k}{k+1}\right) = \ln k - \ln(k+1)$, so $s_n = (\ln 1 - \ln 2) + (\ln 2 - \ln 3) + \cdots + (\ln n - \ln(n+1)) = -\ln(n+1) \to -\infty$. **Answer:** diverges

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11. $\displaystyle\sum_{k=1}^{\infty} \frac{2}{k(k+2)}$

$\frac{2}{k(k+2)} = \frac{1}{k} - \frac{1}{k+2}$. Terms cancel two apart:

$$s_n = \left(1 - \tfrac{1}{3}\right) + \left(\tfrac{1}{2} - \tfrac{1}{4}\right) + \left(\tfrac{1}{3} - \tfrac{1}{5}\right) + \cdots + \left(\tfrac{1}{n-1} - \tfrac{1}{n+1}\right) + \left(\tfrac{1}{n} - \tfrac{1}{n+2}\right) = 1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2} \to \frac{3}{2}.$$

Answer: converges, sum $= \frac{3}{2}$

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12. $\displaystyle\sum_{k=1}^{\infty} \frac{4}{k^2+4k+3}$

$\frac{4}{(k+1)(k+3)} = \frac{2}{k+1} - \frac{2}{k+3}$. Terms cancel two apart:

$$s_n = 2\left[\left(\tfrac{1}{2} - \tfrac{1}{4}\right) + \left(\tfrac{1}{3} - \tfrac{1}{5}\right) + \cdots + \left(\tfrac{1}{n+1} - \tfrac{1}{n+3}\right)\right] = 2\left(\frac{1}{2} + \frac{1}{3} - \frac{1}{n+2} - \frac{1}{n+3}\right) \to \frac{5}{3}.$$

Answer: converges, sum $= \frac{5}{3}$

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13. $\displaystyle\frac{1}{4\cdot 5} + \frac{1}{5\cdot 6} + \frac{1}{6\cdot 7} + \cdots + \frac{1}{(k+3)(k+4)} + \cdots$

This is $\sum_{k=1}^{\infty} \frac{1}{(k+3)(k+4)}$ with $\frac{1}{(k+3)(k+4)} = \frac{1}{k+3} - \frac{1}{k+4}$, so $s_n = \left(\frac{1}{4} - \frac{1}{5}\right) + \left(\frac{1}{5} - \frac{1}{6}\right) + \cdots + \left(\frac{1}{n+3} - \frac{1}{n+4}\right) = \frac{1}{4} - \frac{1}{n+4} \to \frac{1}{4}$.

Answer: converges, sum $= \frac{1}{4}$

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14. $\displaystyle\frac{1}{1\cdot 4} + \frac{1}{4\cdot 7} + \frac{1}{7\cdot 10} + \cdots + \frac{1}{(3k-2)(3k+1)} + \cdots$

This is $\sum_{k=1}^{\infty} \frac{1}{(3k-2)(3k+1)}$ with $\frac{1}{(3k-2)(3k+1)} = \frac{1}{3}\left(\frac{1}{3k-2} - \frac{1}{3k+1}\right)$, so $s_n = \frac{1}{3}\left[\left(1 - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{7}\right) + \cdots + \left(\frac{1}{3n-2} - \frac{1}{3n+1}\right)\right] = \frac{1}{3}\left(1 - \frac{1}{3n+1}\right) \to \frac{1}{3}$.

Answer: converges, sum $= \frac{1}{3}$